

Chapter 6

Lab 6 - Operational Amplifiers Applications

6.1 Objective

The purpose of this lab is to delve into the applications of operational amplifiers. There is an endless number of applications, the possibilities are often not intuitively obvious for students seeing them for the first time.

6.2 Materials

- Laptop with LTSpice
- Analog Discovery
- Breadboard
- Wiring kit
- Lab parts kit with TLV272

6.3 Introduction

There are several standard op amp configurations. While Lab 5 demonstrated the simple inverting and non-inverting cases, this lab expands the scope to include both a summing amplifier and a difference amplifier.

Summing Amplifier

A summing amplifier, Figure 6.1 (c), is effectively the same configuration as an inverting amplifier but has multiple inputs that allow for a weighted output voltage.

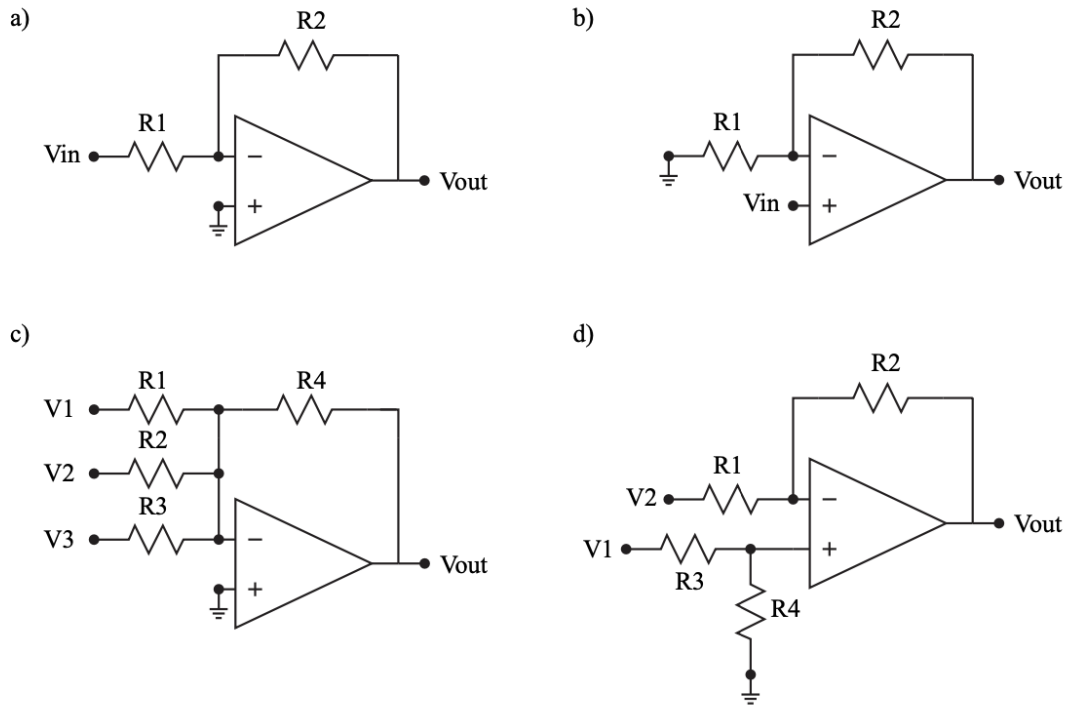


Figure 6.1: Different op amp configurations: inverting amplifier (a), non-inverting amplifier (b), summing amplifier (c), and difference amplifier (d).

$$V_{out} = -\left(\frac{R_4}{R_1}V_1 + \frac{R_4}{R_2}V_2 + \frac{R_4}{R_3}V_3\right) \quad (6.1)$$

When the resistors R_1 , R_2 , R_3 , and R_4 are equal, the output voltage is an equal combination of the input voltages. However, the resistors can be adjusted to assign different weights to the input voltages based on the application.

Difference Amplifier

Difference amplifiers, Figure 6.1 (d), takes the difference between two input voltages, with the output determined by the resistor value ratios.

$$V_{out} = V_1 \left(\frac{R_2}{R_1 + R_2} \right) \left(\frac{R_3 + R_4}{R_3} \right) - V_2 \frac{R_4}{R_3} \quad (6.2)$$

When the resistors $R_1 = R_3$ and $R_2 = R_4$ the output voltage simplifies to:

$$V_{out} = (V_1 - V_2) \frac{R_2}{R_1} \quad (6.3)$$

6.4 Big Picture

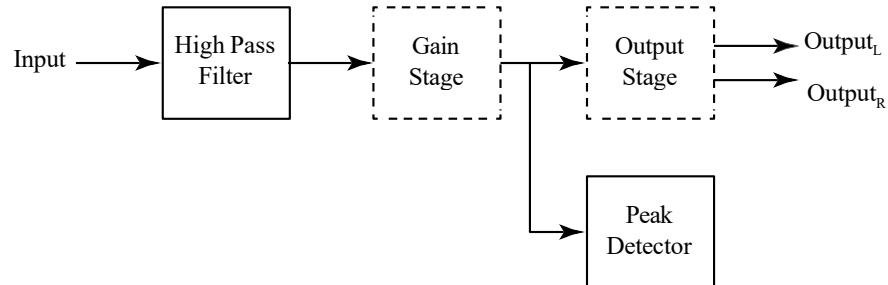


Figure 6.2: Big picture with emphasis on the gain and output stages.

This lab once again focuses on op amps, using them as building blocks to create more complicated circuits. The final project is about as complicated as the final circuit in this lab.

6.5 Pre-Lab Requirements

Complete the following before coming to lab.

6.5.1 Buffering

Buffering is a key op amp application that addresses loading effects, which resistive circuits cannot easily solve. Op amp buffers provide high input impedance and low output impedance, preserving signal integrity across various applications.

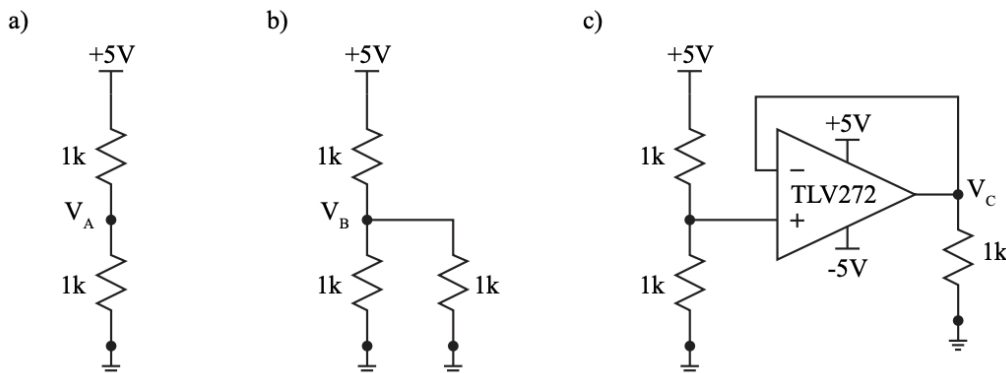


Figure 6.3: A simple voltage divider (a), voltage divider with a load (b), and a voltage divider with a voltage follower between the output and the load (c).

1. Simulate all three circuits in Figure 6.3 using a DC operating point (.op). Keep all the circuits on the same schematic and table the voltages V_A , V_B , and V_C . Table your values and include images of your circuits.
2. Discuss why two of the voltages are approximately 2.5 V but one is not.

6.5.2 Level Shifting

Level shifting in electronic circuits is the process of changing the supply rails from one range to another. For example, when preparing a signal for analog to digital conversion, the signal may be bi-polar (+/-) and must be converted to a single voltage rail. So, instead of a sine wave that swings to +2.5V and -2.5V, 2.5V is added so that the sine wave swings from +5V to 0V with the same amplitude.

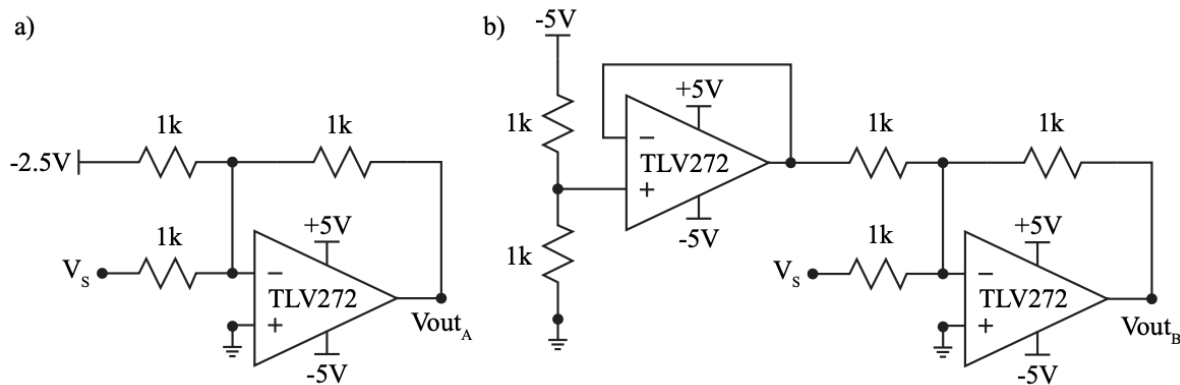


Figure 6.4: A summing amplifier used to add 2.5V to $-V_{IN}$ (a) and the same configuration where the -2.5V source has been replaced with a buffered voltage divider (b).

1. Simulate the circuit in Figure 6.4 (a) using a transient analysis with a stop time of 1m (.tran 1m). Set V_S to a sine wave with a 0 V DC offset, 2 V amplitude, and a 1 kHz frequency. Save an image of the circuit and a plot of the input and output voltage.
2. Explain why the resistor divider in Figure 6.4 (b) uses -5 V instead of 5 V.
3. Practically, a 2.5 V reference may not be available for use and instead must be self-generated. Simulate the circuit in Figure 6.4 (b) using a transient analysis with a stop time of 1m (.tran 1m). Set V_S to a sine wave with a 0 V DC offset, 2 V amplitude, and a 1 kHz frequency. Save an image of the circuit and a plot of the input and output voltage.

4. Construct the circuit in Figure 6.4 (b) on a breadboard. Configure the Wavegen to generate a sine wave at 1 kHz with a 2 V amplitude and a 0 V offset. Plot the input and output on the Scope and save the circuit to demo at the start of lab. If you see clipping at your output, change the three $1\text{k}\Omega$ resistors in the summing op amp to $33\text{k}\Omega$.
5. In the more general case, the full range of input must be accounted for. Simulate the circuit in Figure 6.5 using a transient analysis with a stop time of 1m (.tran 1m). Set V_{IN} to a sine wave with a 0 V DC offset, 5 V amplitude, and a 1 kHz frequency. Save an image of the circuit and a plot of the input and output voltage.

This circuit uses 3 op amps to convert a $\pm 5\text{ V}$ sine wave into a $+5/0\text{ V}$ sine wave. The top op amp generates a -2.5 V offset, while the bottom op amp buffers half of the input voltage. Both outputs are then fed into an (inverting) summing op amp to produce V_{outB} . In short, this circuit adds -2.5 V to half of a $\pm 5\text{ V}$ sine wave, then inverts it to obtain:

$$- [(-5/2) + (-2.5)] = 5\text{ V and } - [(5/2) + (-2.5)] = 0\text{ V}$$

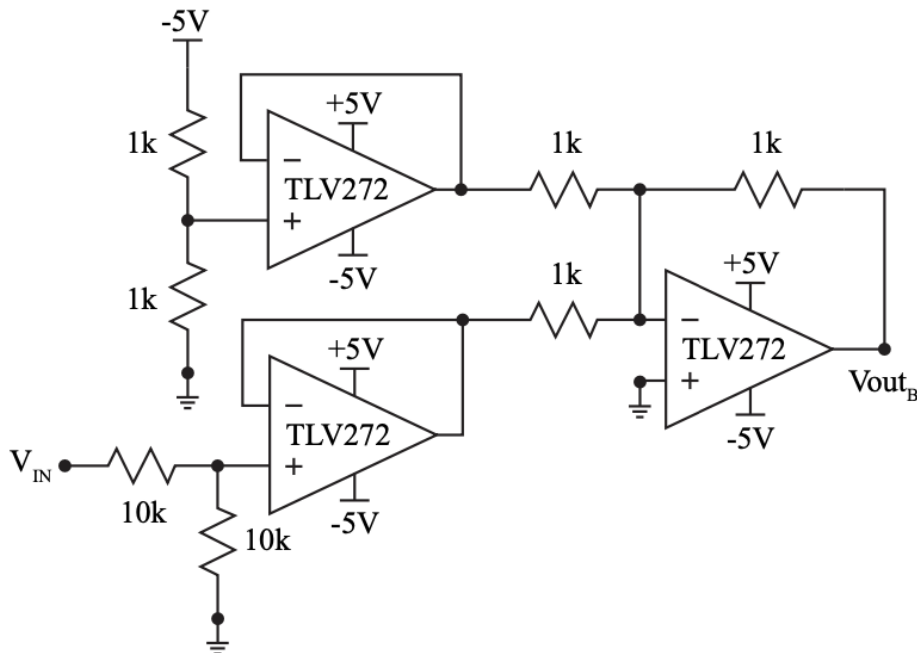


Figure 6.5: A DC level shifter configured to convert the full range of $\pm 5\text{ V}$ to $+5/0\text{ V}$.

6.5.3 Differential to Single Ended Conversion

Differential signals, which are two signals 180 degrees out of phase, offer improved noise performance because any interference affecting both signals in the same way cancels out when they are combined. In other words, only the difference between the signals is

measured, removing unwanted noise. A difference amplifier can be used to convert the differential signal into a single-ended output.

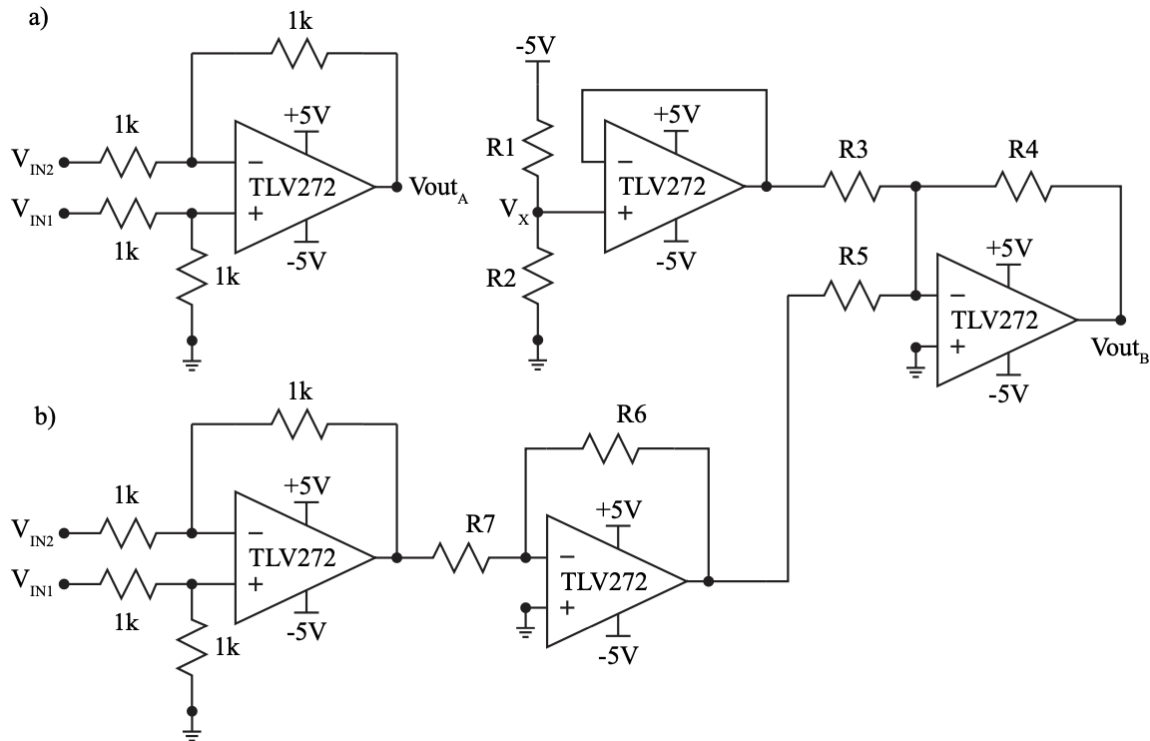


Figure 6.6: A difference amplifier that converts a differential signal to single ended (a) and a combination of the level shifter and differential to single ended converter (b).

1. Simulate the circuit in Figure 6.6 (a) using a transient analysis with a stop time of 1m (.tran 1m). Set V_{IN1} to a sine wave with a 0 V DC offset, 0.5 V amplitude, and a 1k Hz frequency. Set V_{IN2} to a sine wave with a 0 V DC offset, 0.5 V amplitude, a 1k Hz frequency, and a phase of 180 degrees. Save an image of the circuit and a plot of V_{IN1} , V_{IN2} , and V_{out_A} .
2. Determine the resistor values for R_1 through R_7 , recall that there are limited quantities in the lab kit. Choose R_1 and R_2 so that V_X is 2.5 V, R_3 , R_4 , and R_5 so that both voltages are added equally, and R_6 and R_7 so that the output, V_{out_B} utilizes as much of the full range, 0 to 5 V, as possible without clipping. You do not need to use resistors in parallel/series combinations to get an exact value.
3. Simulate the circuit in Figure 6.6 (b) using a transient analysis with a stop time of 1m (.tran 1m). Set V_{S1} and V_{S2} to their previously defined values. Save an image of the circuit and a plot of V_{IN1} , V_{IN2} , and V_{out_B} .
4. Explain how the circuit in Figure 6.6 (b) works in a few sentences, discussing the purpose of each op amp.

6.5.4 Pre-Lab Checklist

The following must be completed at the start of lab.

1. Section 6.5.1
 - (a) Item 1: Image of the circuit and a table of output voltages.
 - (b) Item 2: Answer to question.
2. Section 6.5.2
 - (a) Item 1: Image of the circuit and a plot of the input and output voltages.
 - (b) Item 2: Answer to question.
 - (c) Item 3: Image of the circuit and a plot of the input and output voltages.
 - (d) Item 4: Plot of the input and output voltages and a circuit demonstration.
 - (e) Item 5: Image of the circuit and a plot of the input and output voltages.
3. Section 6.5.3
 - (a) Item 1: Image of the circuit and a plot of the input and output voltages.
 - (b) Item 3: Image of the circuit and a plot of the input and output voltages.
 - (c) Item 4: Answer to question.

6.6 In-Lab Requirements

Complete the following in lab.

6.6.1 Construction

1. Construct the circuit from Section 6.5.2 Item 5. Set Wavegen 1, V_{IN} to a sine wave with a 5 V amplitude, 0 V offset, and a frequency of 1 kHz.
2. Plot the input and output voltages, save an image of the plot.
3. Construct the circuit from Section 6.5.3 Item 3. Set Wavegen 1 to V_{IN1} and Wavegen 2 to V_{IN2} . Synchronize Wavegen 1 and Wavegen 2 using the dropdown in the wavegen tab of waveforms. It is located two tabs to the right of "Run all" and is defaulted to "No synchronization".
4. Plot the input (differential) and the output voltages, then save an image of the plot. Typically, scope wire 2 (with the white stripe) is connected to ground because scopes

measure voltages relative to another signal. In this case, however, you connect scope wire 1 to Wavegen 1 and scope wire 2 to Wavegen 2, allowing the scope to measure the difference between them.

6.7 Write Up

Include the following in the write up.

1. Section 6.6.1
 - (a) Item 1: LTspice schematic of the circuit.
 - (b) Item 2: Plot of the input and output voltages.
 - (c) Item 3: LTspice schematic of the circuit with all calculated resistors.
 - (d) Item 4: Plot of the input and output voltages.

Discuss the two circuits constructed in lab, DC level shifter and differential to single ended converter. Explain how each circuit works at the block level and how resistor tolerance could or could not influence the output. Also explain the design choices for the circuit, Section 6.5.3 Item 3, constructed in lab. Which resistors were chosen and why?